

JavaScript NaN

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Invalid Number Operations

In JavaScript, `NaN` is short for **Not a Number**. `NaN` is a JavaScript number that is not a legal number.

You get `NaN` when JavaScript cannot calculate a number.

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript NaN</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Invalid Number Operations</h4>
<p id="demo"></p>

<script>
let x = 100 / "Apple";

document.getElementById("demo").innerHTML = x;
</script>

</body>
</html>
```

Example 1

Result [View Example](#)

NaN is a Number

The type of `NaN` is `number`. This may look strange, but `NaN` belongs to the JavaScript number type.

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript NaN</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>NaN is a Number</h4>
<p id="demo"></p>

<script>
let x = NaN;

document.getElementById("demo").innerHTML = typeof x;
</script>

</body>
</html>
```

Example 2

Result [View Example](#)

Numeric Strings

JavaScript tries to convert numeric strings to numbers in arithmetic operations.

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript NaN</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Numeric Strings</h4>
<p id="demo"></p>

<script>
let x = 100 / "10";

document.getElementById("demo").innerHTML = x;
</script>

</body>
</html>
```

The result is `10` , because `"10"` is converted to the number `10` .

Example 3

Result [View Example](#)

Non-Numeric Strings

A non-numeric string cannot be converted to a number.

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript NaN</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Non-Numeric Strings</h4>
<p id="demo"></p>

<script>
let x = 100 / "Apple";

document.getElementById("demo").innerHTML = x;
</script>

</body>
</html>
```

The result is `NaN` , because `"Apple"` cannot be converted to a number.

Example 4

Result [View Example](#)

Using isNaN()

You can use the JavaScript function `isNaN()` to find out if a value is not a number.

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript NaN</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Using isNaN()</h4>
<p id="demo"></p>

<script>
let x = 100 / "Apple";

document.getElementById("demo").innerHTML = isNaN(x);
</script>

</body>
</html>
```

Example 5

Result [View Example](#)

NaN is Not Equal to Itself

NaN is the only JavaScript value that is not equal to itself.

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript NaN</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>NaN is Not Equal to Itself</h4>
<p id="demo"></p>

<script>
let x = NaN;

document.getElementById("demo").innerHTML = x == x;
</script>

</body>
</html>
```

To test for `NaN` , use `isNaN()` .

Example 6

Result [View Example](#)

NaN in Math

If you use `NaN` in a mathematical operation, the result will also be `NaN` .

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript NaN</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>NaN in Math</h4>
<p id="demo"></p>

<script>
let x = NaN;
let y = 5;

document.getElementById("demo").innerHTML = x + y;
</script>

</body>
</html>
```

Example 7

Result [View Example](#)

Note

NaN means **Not a Number**. But the type of NaN is number . Use isNaN() to check if a value is NaN .

Document

Document in project

You can [Download PDF](#) file.

Reference

- [W3Schools JavaScript NaN](#)